UNSW MARKETING ANALYTICS HACKATHON 2026

Getting slow loans funded faster

What 1.45 million Kiva loans say about predicting slow loans and supporting them

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UNSW SYDNEY





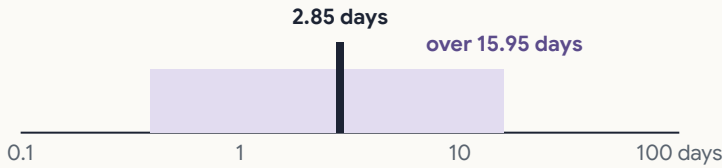


Act on predictable delays now; test writing advice before giving it

Problem, solution and proof across all evidence layers

THE PROBLEM

2.85 days

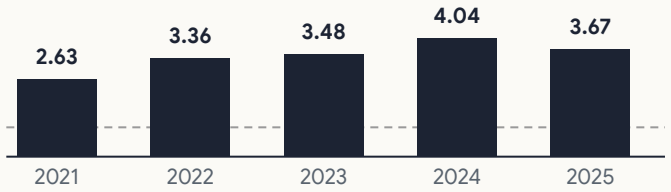


Median wait to full funding

A quarter of loans wait over 15.95 days.

OUR SOLUTION

2.6–4.0×

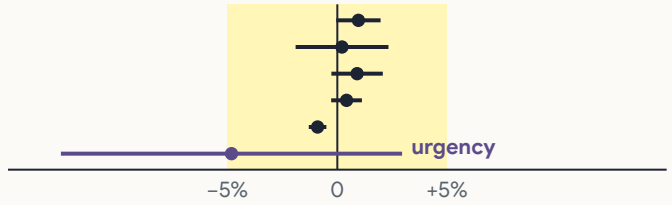


Better than chance at flagging slow loans

Loan details flag the slowest fifth early, so support can go there first.

THE PROOF

±5%



Framing links stay small

Five framings stay within ±5% of funding time, so rewriting stories is not the lever.

Our ask to Kiva: pilot a slow-funding flag with lender-side support, and test one truthful prompt.

Predict Prioritise Test

For whom and when: no period, loan size, gender or region shows a large framing effect; a few rare-word patterns are hypotheses to test.

Agenda

01

The problem

Who waits, and where?

- How a loan gets funded 4
- The wait 5
- Where loans and delays sit 6

02

What drives the wait

What does the evidence show?

- What we read in each loan 7
- Key insight 8
- Four analyses 9
- Country uncertainty 10
- Programme controls 11
- Why not rewrite stories 12
- For whom and when 13
- What text adds 14
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03

Our solution

What can Kiva do now?

- Predict, prioritise, test 16
- Predict 17
- Prioritise 18
- Test 19

04

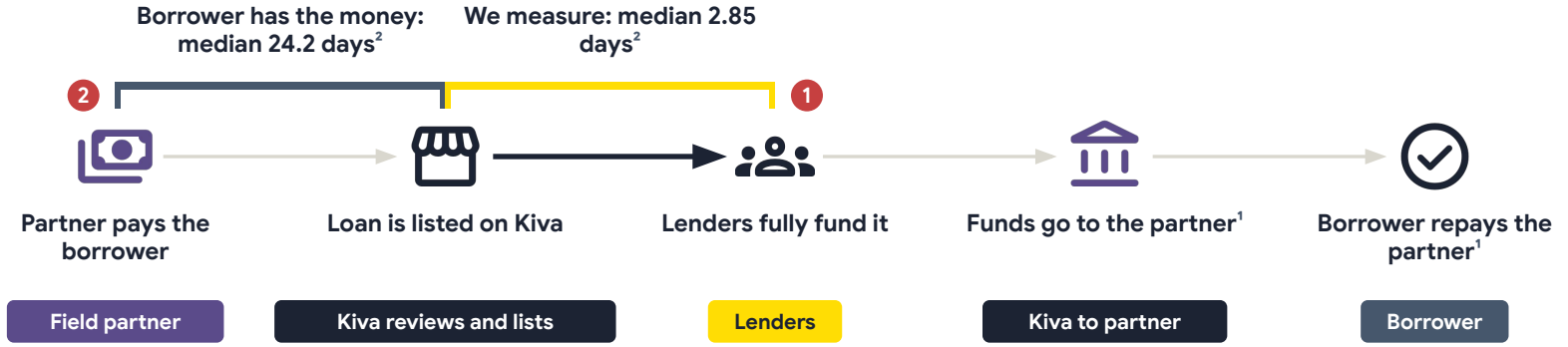
Safe next steps

What keeps it honest, and what comes next?

- Guardrails 20
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Borrowers usually have the money before lenders fund the loan

Descriptive: how a typical pre-disbursed Kiva loan moves, who acts at each stage, and what we measure



Loans disbursed to the borrower before listing



Who is in the data

In the data: 1,453,840 completed loans, including 1,637 refunded

Not in the data: listings that expired or never funded 4

Figure 1. Journey of a typical pre-disbursed Kiva loan, not to scale; medians from the full-data audit.

[1] Typical field-partner flow; these steps are not observed in the data.

[2] Separate medians: 24.2 days covers the 96.4% disbursed before listing, and the two do not add up along one loan.

Scope: 1,453,846 input records; 6 negative durations excluded; 1,637 refunded records included.

1 We study the lender-side step

Our outcome runs from **listing to full funding**, median 2.85 days.

2 Borrowers usually already have the money

96.4% of loans reach the borrower a median **24.2 days** before listing.

3 Speed turns over partner capital

Only **3.6%** are paid out after listing, so slow funding mainly holds up the partner's next loans.

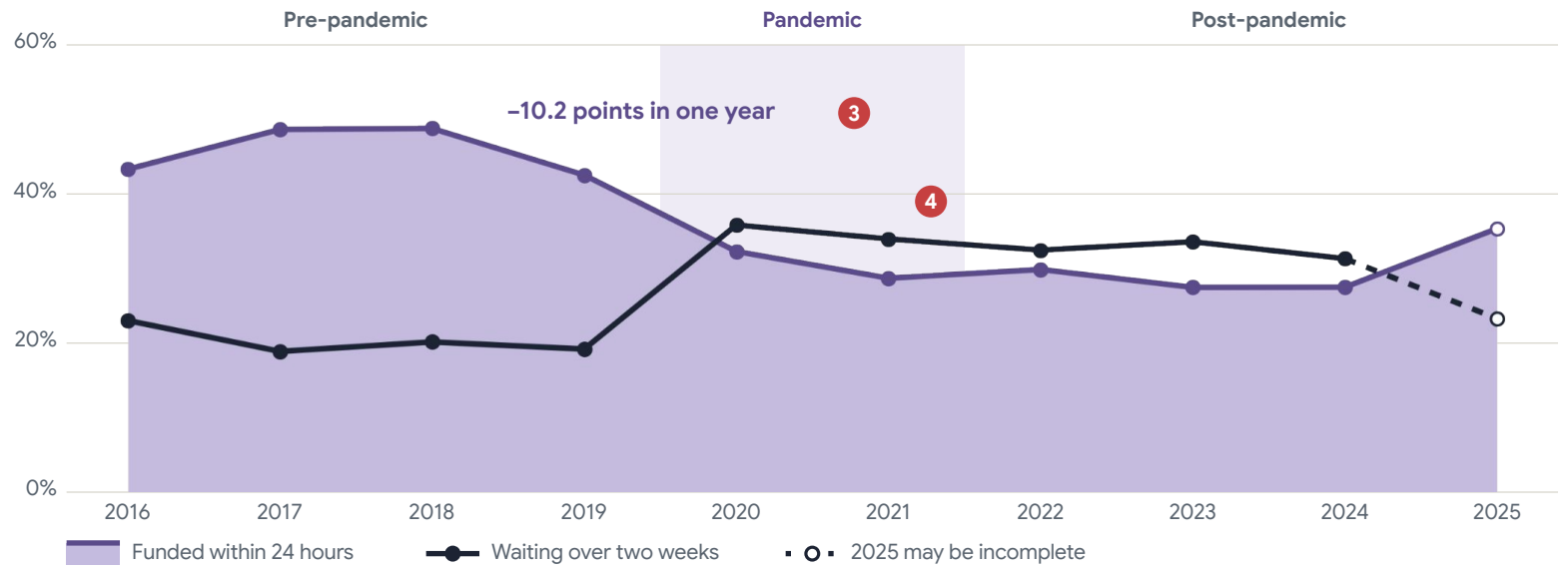
4 Unfunded loans are missing

Only loans that **eventually funded** are in the data; expired listings are not.

→ Why speed is a lender-side problem

Waits got longer after 2019: a third of loans now wait over two weeks

Descriptive: platform funding time from listing to full funding, 1,453,840 completed loans, by posting year



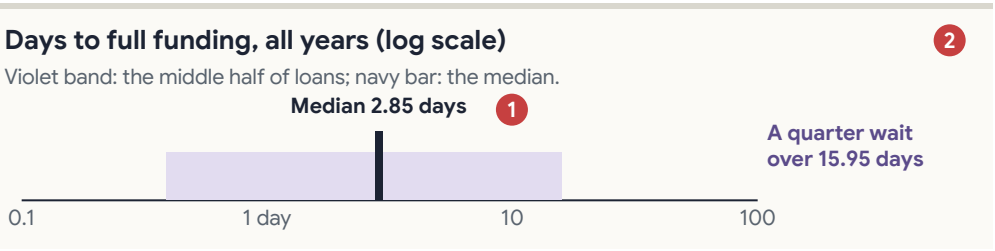




Figure 2. Same-day share and share waiting over two weeks by posting year; funding-time quartiles, all years.

1 Most loans fund within days

Half are fully funded within **2.85 days**, a quarter within **0.39 days**.

2 A long tail waits weeks

A quarter of loans wait **over 15.95 days**, while borrowers already have the money.

3 Same-day funding fell in one year

From **42.5% in 2019 to 32.3% in 2020**, a fall of 10.2 points.

4 Long waits became common

Loans waiting over two weeks rose from **19.2% to 35.8%**; in 2021 that was **55,408 loans**.

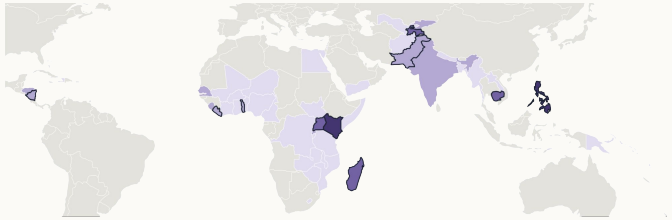
→ Why early warning matters

Over half of all two-week waits come from just three countries

Descriptive: where loans sit and where they wait over two weeks, by country, 1,453,840 completed loans

(a) Where the loans are

Ten of 48 countries hold 81% of loans (outlined on the map); table share cells use the map's colour scale.



	Country	Region	Loans	Share	Same-day
1	**Philippines**	Asia	521,586	35.9%	48%
2	**Kenya**	Africa	266,562	18.3%	29%
3	**Uganda**	Africa	69,651	4.8%	31%
4	**Tajikistan**	Asia	68,865	4.7%	18%
5	**Cambodia**	Asia	58,074	4.0%	49%
6	**Madagascar**	Africa	55,670	3.8%	67%
7	**Pakistan**	Asia	42,767	2.9%	15%
8	**Nicaragua**	Central America	38,520	2.6%	15%
9	**Liberia**	Africa	28,847	2.0%	28%
10	**Togo**	Africa	27,513	1.9%	38%

(b) Where the long waits are

Each bar is all of a country's loans, split by how long they took to fund.

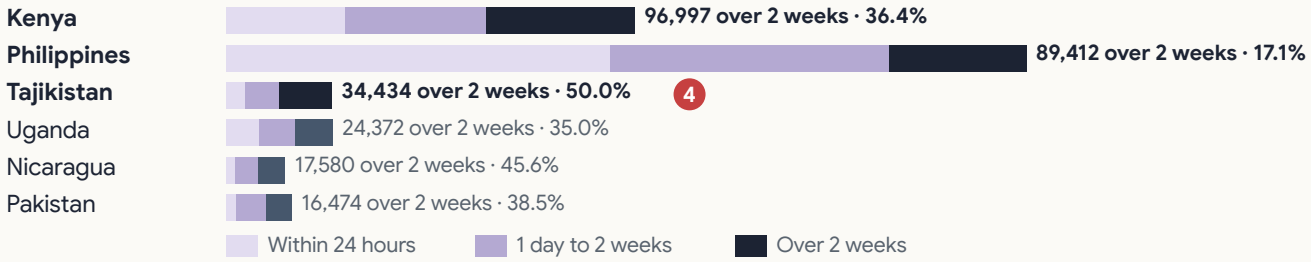


Figure 3. Share of all loans by country (a, map and table) and loans waiting over two weeks (b); same-day: funded within 24 hours.

1 Ten countries hold most loans

Ten of 48 countries hold **81% of loans**; Asia and Africa hold **92.8%**.

2 Speed varies widely by country

Same-day funding runs from **15% in Pakistan to 67% in Madagascar**.

3 Three countries hold half the long waits

Kenya, the Philippines and Tajikistan hold **56.1%** of the **393,847** loans that waited over two weeks.

4 Large markets wait too

In Kenya **36.4%** of loans wait over two weeks; in Tajikistan **50.0%**.

→ Why the flag must work country by country

Each loan story is scored on eight framing measures

Illustrative listing: what is fixed when a loan is listed, and what we measure in the description



FIXED AT LISTING 1

Amount, median	Term, median	Borrowers, median
US$325	**12 months**	**1**
Sector, most common	Country, most common	Partner
Agriculture	**Philippines**	**Field partner**

EDITABLE DESCRIPTION 4

> She has a banana stall at the market to support her family. She wants to buy more stock quickly, before prices rise. She is thankful to every lender who helps her business grow.

The eight framing measures, counted per 100 words, with the share of loans using each (plus a sentiment score) 2

- **Family** — Relatives and schooling — e.g. children · in 88.7%
- **Business** — Trade and stock words — e.g. market · in 85.0%
- **Basic needs** — Water, health, housing — e.g. water · in 23.3%
- **Agency** — Planning and self-direction — e.g. decided · in 54.1%
- **Gratitude** — Thanks and blessings — e.g. thankful · in 19.6%
- **Urgency** — Time-pressure words — e.g. quickly · in 1.3% 3
- **First person** — The borrower's own voice — e.g. my · in 1.0%
- **Third person** — Written about the borrower — e.g. her · in 97.3%

Figure 4. Illustrative composite listing, not a real borrower; dataset medians and most common categories. Word lists are not yet validated (gate 1).

1 Most of the listing is fixed

Amount, term, sector, country and partner are set; the **description** is the part a partner writes.

2 We score eight framing measures

Counted with **word lists per 100 words**, plus VADER sentiment and text form.

3 Some measures are rare

Urgency appears in **1.3%** of descriptions and first person in **1.0%**; only **11.3%** never mention family.

4 Pronouns mostly reflect the writer

Descriptions are **written about the borrower**, so third person is context.

→ Why wording is tested, not assumed

Slow funding is a lender-side delay that loan details predict, not wording

Three findings across the outcome audit, the early-warning check and the equivalence test, each leading to a step of the solution

The wait is on the lender side

1

96.4%

of loans reach the borrower before listing



→4

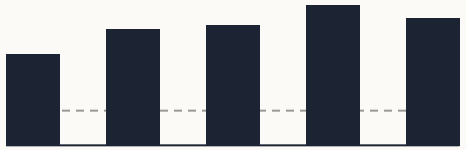
Leads to **Prioritise**

Loan details predict who waits

2

2.6–4.0×

better than chance at flagging the slowest fifth



→17

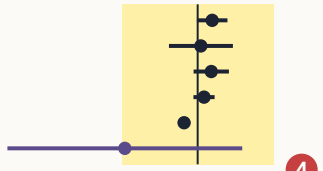
Leads to **Predict**

Wording makes little difference

3

±5%

five framings stay within this share of funding time



→12 →13

Leads to **Test first**

1 Act on the lender side

96.4% of loans reach the borrower before listing, so the wait ties up partner capital, not the borrower.

2 Predict slow loans early

A boosted tree on loan structure flags the slowest fifth **2.6–4.0× better than chance**.

3 Don't chase framing tweaks

Five framings stay **within ±5%** of funding time, and text adds little to the flag.

4 It holds in every group

Six framings stay small in **all 10 subgroups** of period, loan size, gender and region.

→ The evidence behind each insight follows

Four analyses feed the solution; none is a causal claim

One dataset: the equivalence test builds on the robustness checks, and the forecast benchmark asks a separate question

Kiva loans, 2016–2025: 1,453,840 valid completed durations in 48 countries 1

Explanatory models

Which features are linked to funding speed?

64 of 120 significant terms lost significance

Feeds Test 2

Robustness checks

Do the links survive richer controls?



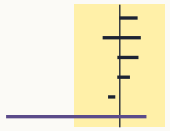
1 of 36 passes

One tiny link passes, in one model only

Feeds Prioritise 3

Equivalence test

Which links are small enough to rule out a large effect?



Five framings small; urgency unresolved

Feeds Prioritise

Forecast benchmark

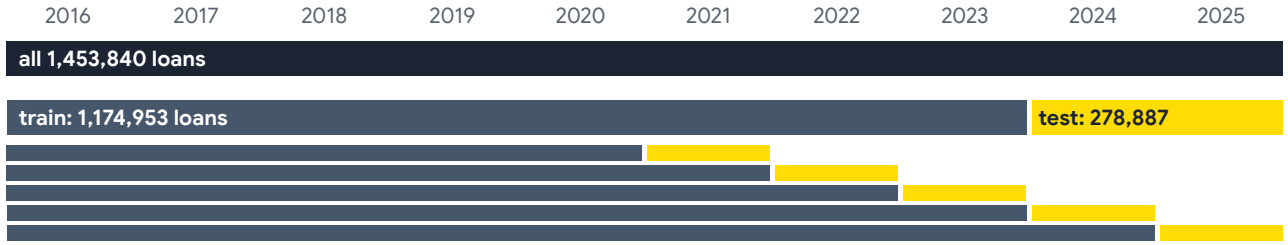
Does text improve a forecast built on loan structure?



Small gain: 0.26–0.39 days, about half context

Feeds Predict 4

Years each analysis uses



1 One dataset, four questions

All four use the same **1,453,840 loans**, and each supports a different kind of claim.

2 Uncertainty comes first

Clustering by 48 countries removed **64 of 120** significant terms.

3 Framing signals stay small

After a wild cluster bootstrap, **1 of 36** tests passes: third person, in one model only.

4 Forecasting is tested separately

Each model trains only on **loans completed before its test year**.

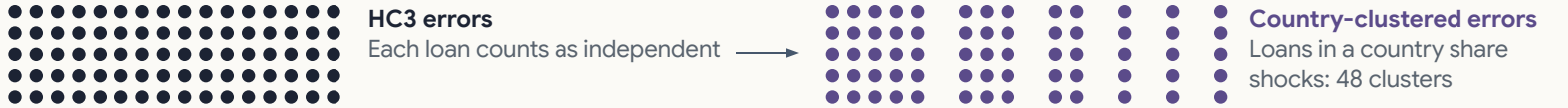
→ None is causal, so the solution ends with a test

Clustering by country removed 64 of 120 significant terms

Explanatory models: HC3 versus 48-country clustered errors on 1,453,840 loans

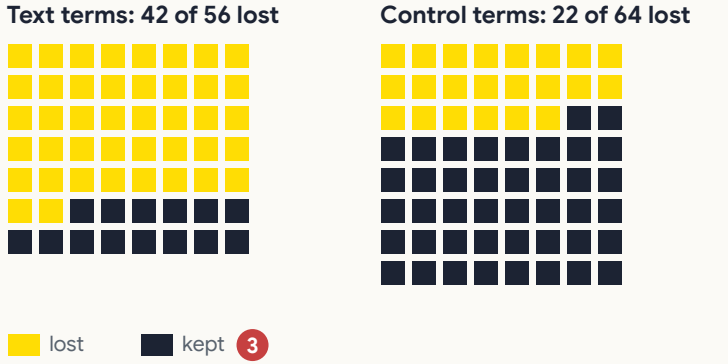
Why the errors widen

The loans and estimates stay the same; only the assumption about how loans vary together changes.



Which terms lost significance

Each square is one term significant under HC3.



Urgency before and after

p-value for urgency in each model; clustering moves both past the 0.05 line.

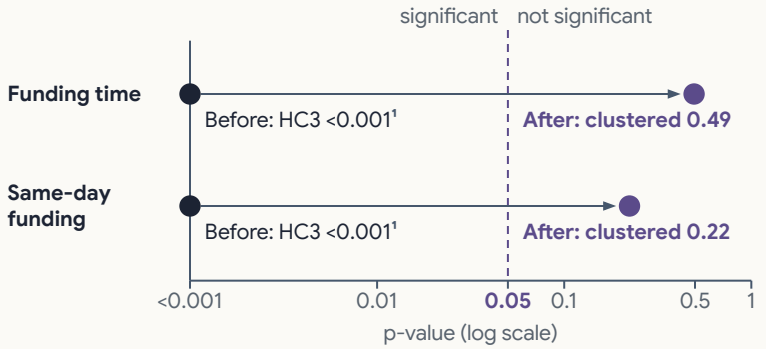


Figure 5. Explanatory models, HC3 versus errors clustered by country (48 clusters). Dots in the top panel are illustrative.

[1] Printed as 0.0000 in the saved run. Text terms include interactions of a text measure with a control.

1 Few countries dominate

Ten countries hold **81% of loans**; regional patterns rest on **two countries each**.

2 Text terms are hit hardest

42 of 56 text terms lost significance, against 22 of 64 controls.

3 Few text terms survive

Only **14** stay significant: sentiment in both models and 12 family-framing interactions.

4 Urgency no longer passes

Its p-value rose from below 0.001 to **0.49** for funding time and 0.22 for same-day funding.

→ Why the flag is checked country by country

One tiny signal passes the strictest test, but not with programme controls

Robustness checks: programme controls, then a wild cluster bootstrap across 48 countries

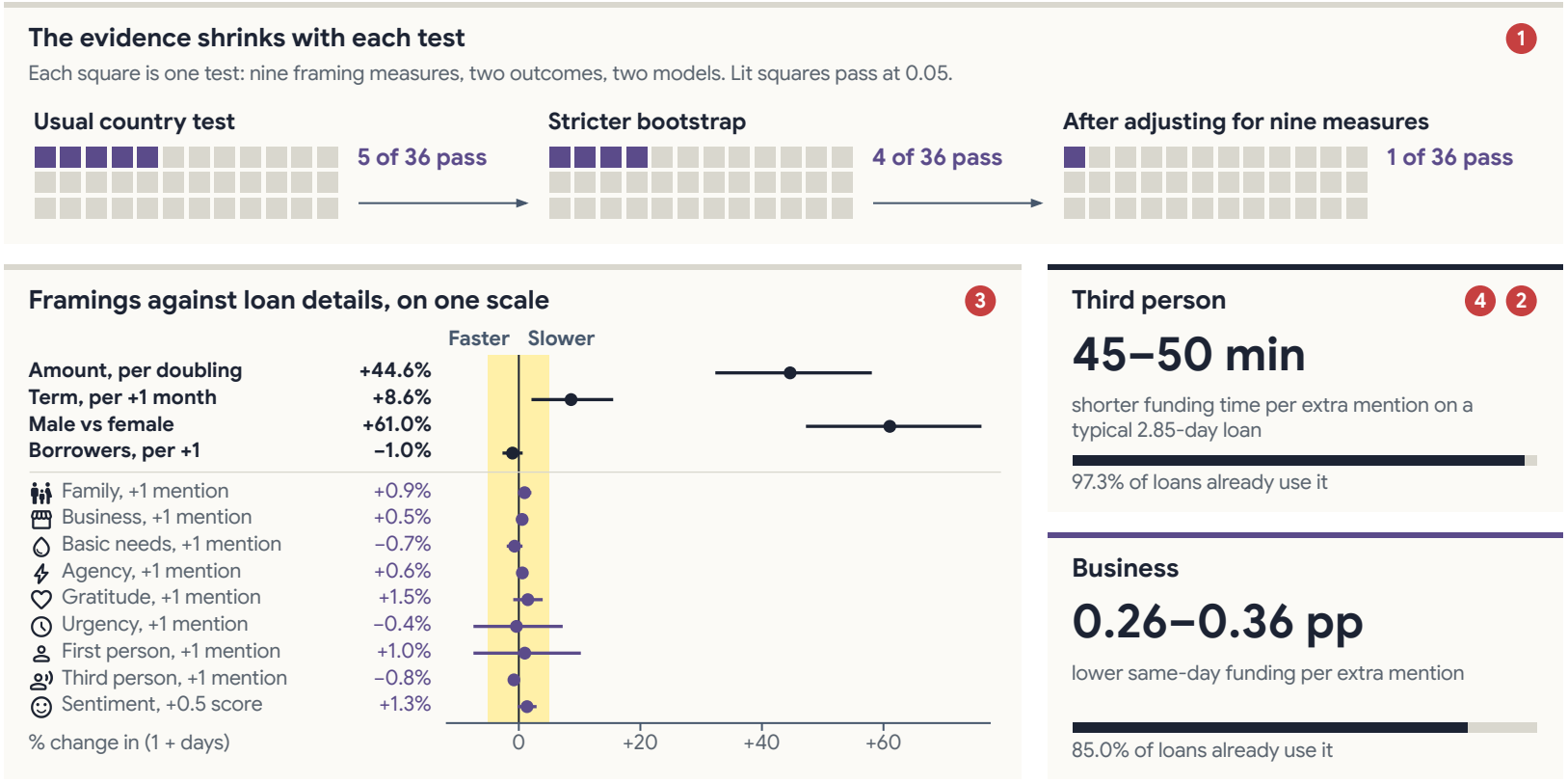


Figure 6. Per extra mention per 100 words; BH-adjusted wild cluster bootstrap p across nine terms (Webb weights, 9,999 draws). Typical loan: the 2.85-day median. **Scale.** Change in (1 + days), not days, with 95% country-clustered ranges; programme-adjusted model. Amount: a doubling from the US$325 median, within its size band.

1 The evidence shrinks with each test

5 of 36 tests pass the usual country test, 4 the bootstrap, **1** after adjustment.

2 The one that passes is tiny

Third person, expanded model: **−0.9%** per mention (BH p 0.008); with programme controls, BH p 0.426.

3 Loan details are on another scale

A doubled amount goes with **+44.6%** and a month of term **+8.6%** in (1 + days); framings stay near ±1.5%.

4 Pronouns may reflect the writer

Third person is in **97.3%** of loans, stays within ±2%, and partners usually write the text.

→ Why we prioritise loans, not wording

Five framings have small links to funding time; urgency remains unknown

Equivalence test of the robustness-check coefficients; exploratory, bound set after the estimates

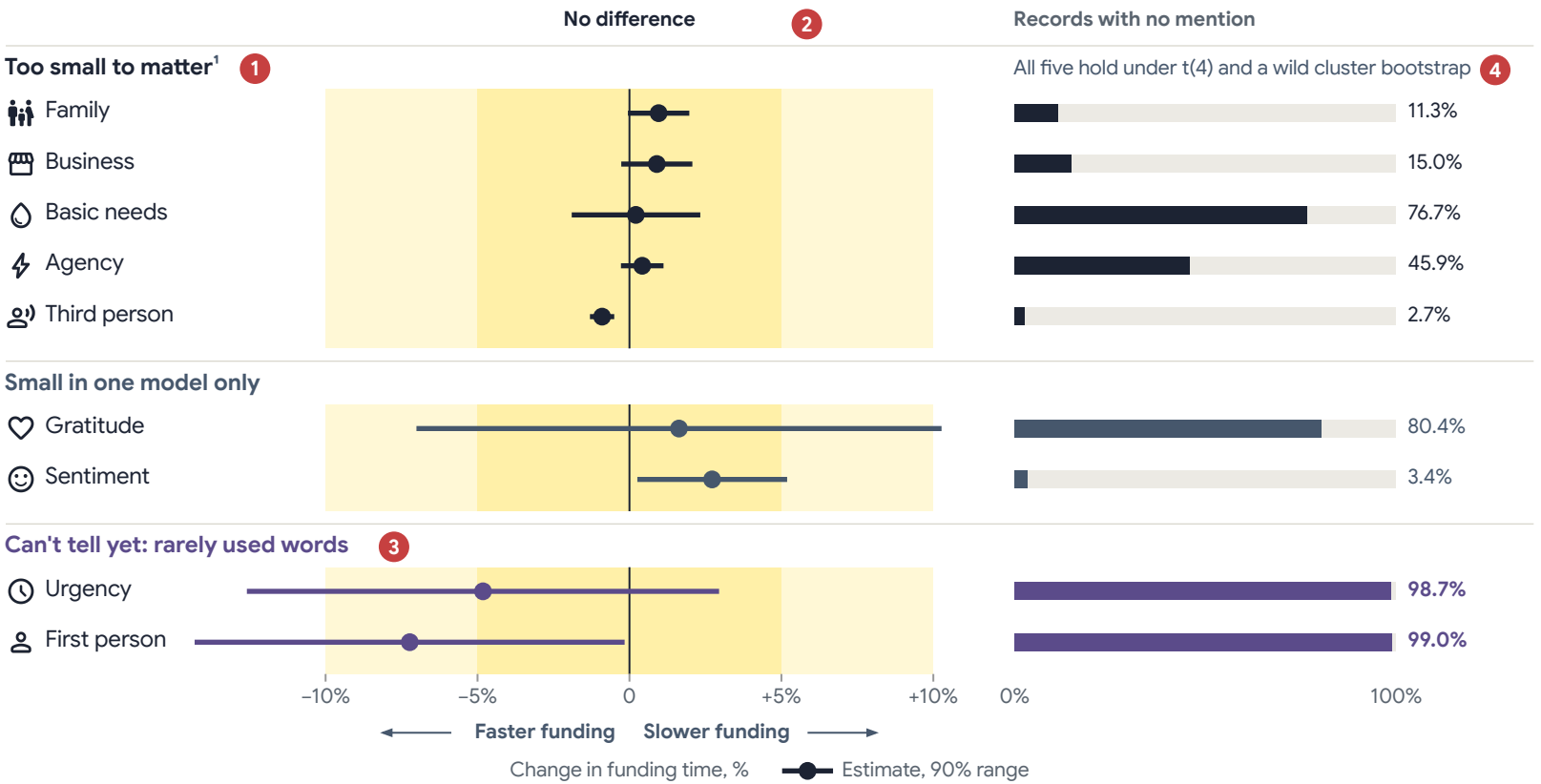


Figure 7. Change in funding time per extra mention per 100 words (sentiment: per +0.5 score), expanded model, 1,453,840 loans.

[1] Within ±5% of funding time: ±0.05 on log(1 + days), BH-adjusted; bound set after the estimates.

1 Five framings have small links

Family, business, basic needs, agency and third person stay **within ±5%** in both models.

2 Small means under five hours

±5% is about **4.5–4.7 hours** on a typical 2.85-day loan.

3 Urgency is too rare to judge

Under **2% of descriptions** use urgency or first person; their ranges still reach **6–14%**.

4 The verdicts hold under stricter tests

A wild cluster bootstrap keeps **35 of 36 verdicts**; three framings stay small even within **±2%**.

→ Why we don't chase framing tweaks

Six framings stay small in every period, loan size, gender and region

Exploratory: programme-adjusted model refitted in 10 subgroups, with the same ±5% bound as the last slide

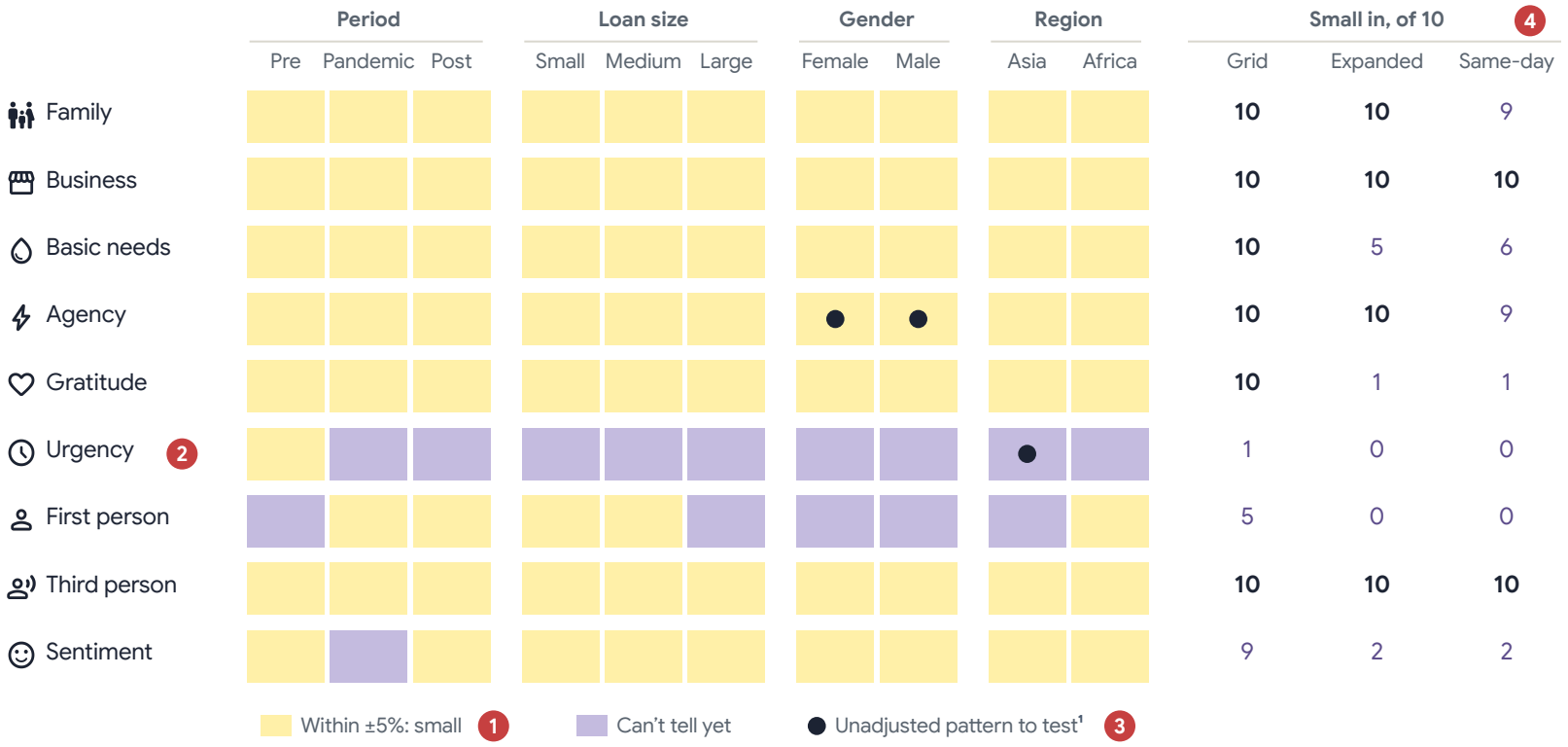


Figure 8. Funding-time verdict by framing and subgroup, programme-adjusted model, 1,453,840 loans. Expanded: funding time. Same-day: ±1 pp.

[1] Unadjusted p-values among about 180 subgroup tests; hypotheses for the Test step, not findings.

1 Small everywhere, not just on average

The five framings on the last slide plus gratitude stay within ±5% in **all 10 subgroups**; no family interaction with period or loan size survives clustering.

2 Rare words stay open in most groups

Urgency is small only before the pandemic (**1 of 10**); first person in **5 of 10**.

3 Two patterns to test, not to act on

Urgency in Asia and agency by gender stand out among about **180** unadjusted tests.

4 The count depends on the model

In the expanded model **4** framings stay small in all 10; for same-day funding, **2**.

→ Why one playbook serves every group

Text adds a small, inconsistent forecast gain; about half is context

Forecast benchmark: gain from adding text to loan details, five annual test years, identical rows

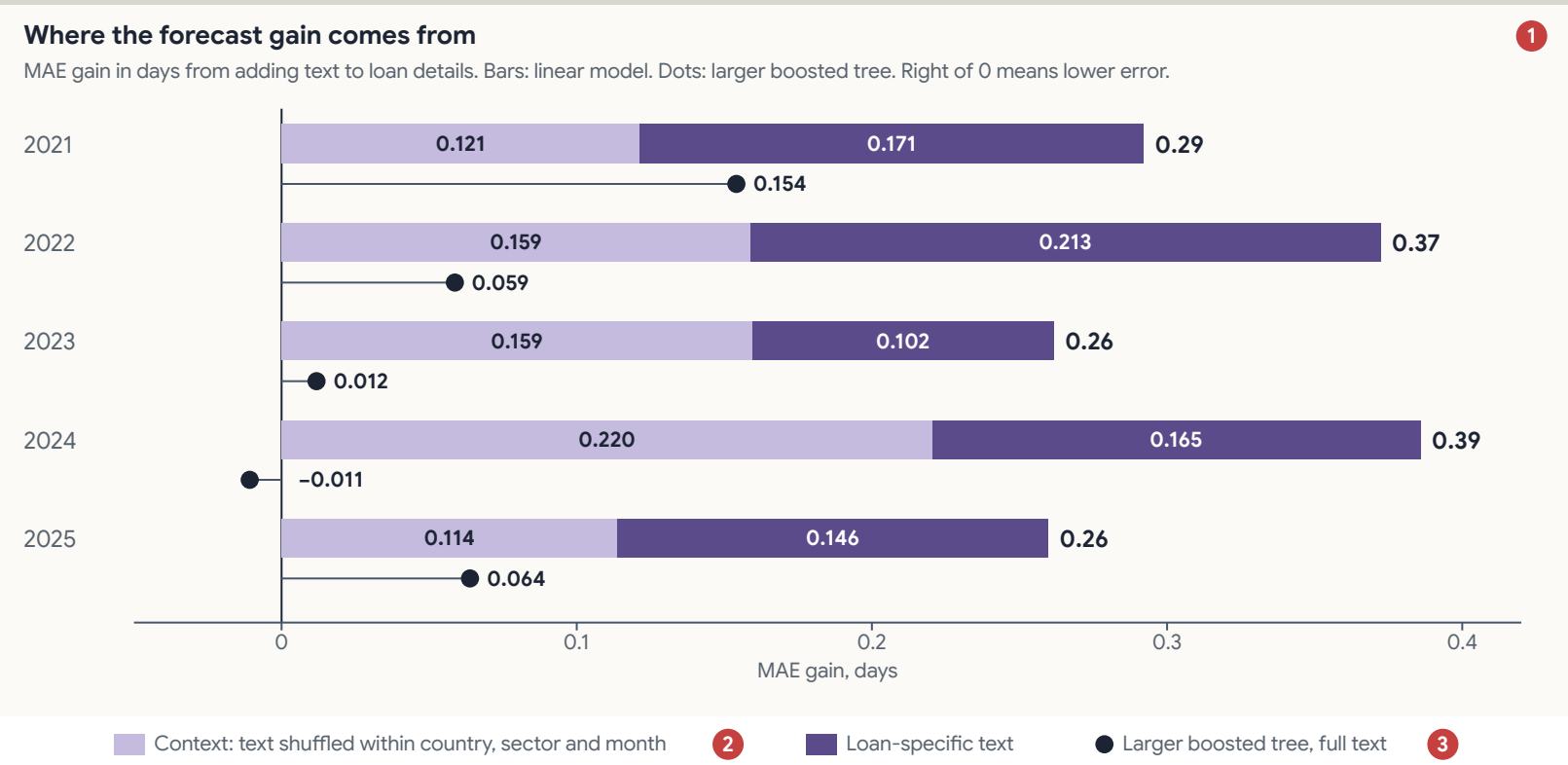


Figure 9. Annual MAE gain in days, identical rows; bars from the placebo refit, within 0.003 days of the benchmark. Text = form, word lists, VADER and TF-IDF.

1 A small gain on a simple model

Text, mainly **TF-IDF word patterns**, lowers a linear model's error by 0.26–0.39 days, about 7 hours.

2 About half is context

Shuffled text keeps **42–61%** of the gain; the loan-specific part is 0.10–0.21 days.

3 A stronger tree gains less

A larger boosted tree gains **at most about 0.15 days**, and under 0.07 in four of five years.

4 It does not show rewriting helps

Text alone does worse than loan details, and no forecast tests an edit.

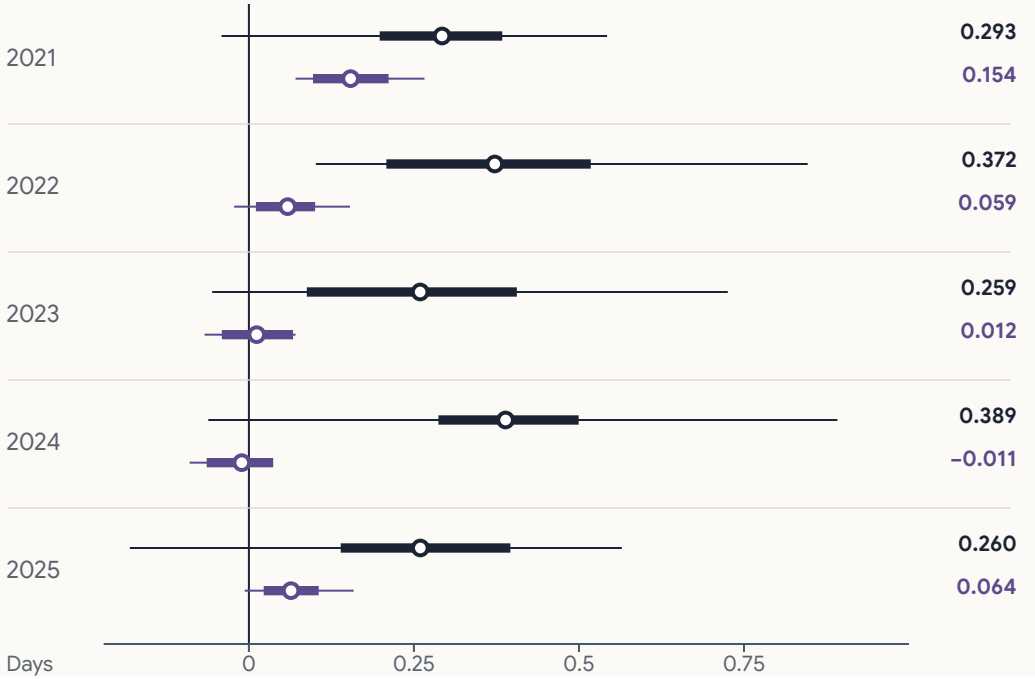
→ Why we predict with loan details first

The text gain is fragile across countries and on a stronger model

Forecast benchmark: annual MAE gain in days, with 95% ranges from resampling months or countries

The text gain by year: two models, two ranges

MAE gain in days from adding text to loan details; right of 0 means lower error. The dot is the gain.



Linear model | Thick: month range (1) | Larger boosted tree (2) | Thin: country range (3)

How often the gain clears zero

Years out of 5 where the 95% range excludes 0. More years means a steadier gain.

By month | By country

Linear model
Holds over time, not across countries

5 ●●●●● → 1 ○●○○○

Larger boosted tree
Less consistent even over time

3 ●●○○● → 1 ●○○○○

Loan-specific part of the text gain
The loan's own words; still uneven

4 ●●○●● → 2 ○●○○●

● Range excludes zero ○ Includes zero

Figure 10. Annual MAE gain in days from adding text, identical rows. Text = form, word-list counts, VADER sentiment and TF-IDF word patterns.

[1] Forecasts capped at the longest training wait (173.4 days); a diagnostic, not the reported model. (4)

1 Stable across months, linear only

The linear gain clears zero **every year** when months are resampled.

2 A stronger tree gains little

Its largest yearly gain is **about 0.15 days**, against 0.26–0.39 on the linear model.

3 Across countries it is fragile

Few years clear zero once countries are resampled, even for the loan-specific part.

4 Extreme forecasts explain part of it

With forecasts capped[1], the linear gain's country range includes zero in only **1 of 5** years.

→ Why the flag needs validating by country

Our solution: predict slow loans, prioritise support, test before advising

A three-step framework for Kiva and partners, built only on what the evidence supports

1 Predict

What
Score each listing's risk of slow funding when it is posted.

Who
kiva platform

When
At posting

Status
Retrospective check only

Evidence: Slide 17

2 Prioritise

What
Send lender-side support to flagged loans.

Who
kiva and partners

When
While the loan raises funds

Status
Needs a pilot

Evidence: Slides 12–13 and 18

3 Test

What
Earn the right to give writing advice.

Who
Research with partners

When
Before any writing advice

Status
Not started

Evidence: Slide 19

Independent hackathon analysis; not affiliated with or endorsed by Kiva.

1 Start with what is known at posting

A listing's risk can be scored the moment it is posted, **from loan details**.

2 Act on loans, not wording

Support goes to flagged loans, because **most framing links stay small**.

3 Advice comes last

No description edit has been tested, so **writing advice waits for a pilot**.

4 Nothing is deployed yet

The flag is checked on past loans only, and the **support levers still need a controlled pilot**.

→ Why we start by flagging slow loans

A flag at listing beats a simple historical lookup in every year

Concept to pilot: retrospective check on completed loans only, five annual test years

Concept to pilot

4

Listing at posting time

Agriculture · US$325 · 12-month term

Illustrative listing, not a real borrower

Predicted risk of slow funding

top 10%

Flag: likely slow to fund

Chance of same-day funding: low

Suggested support

- Feature it to lenders
- Offer matching funds
- Alert lenders who follow the sector
- Ask the partner to review it

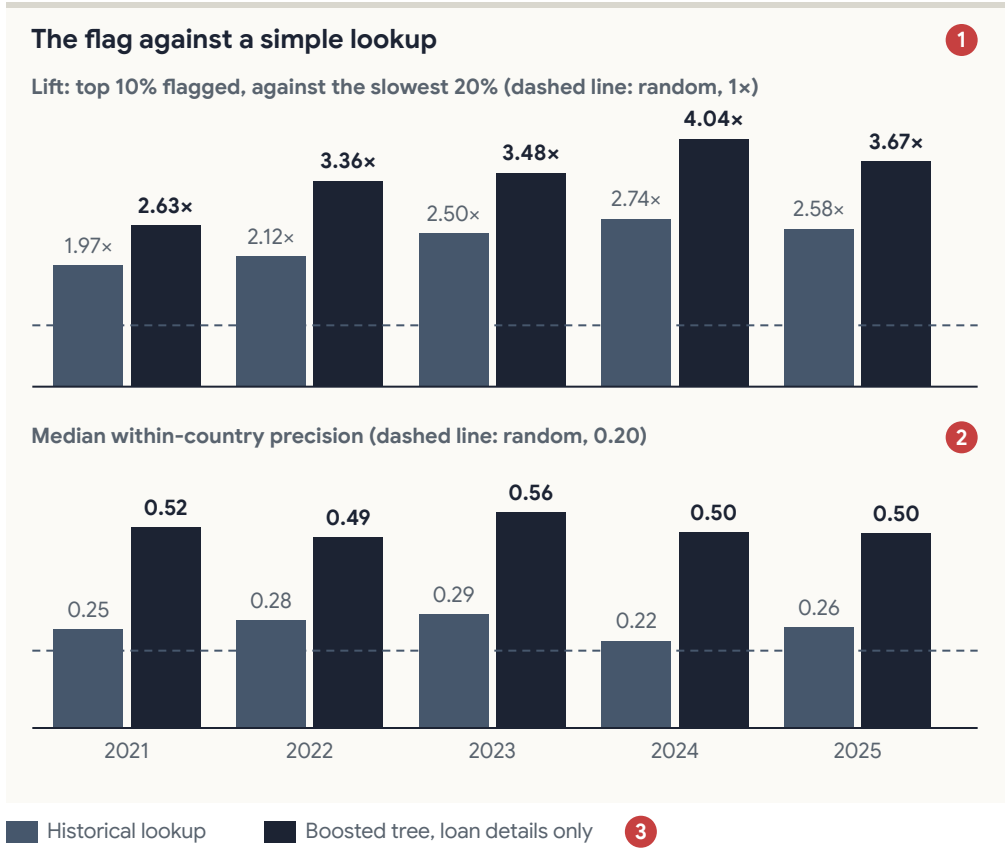


Figure 11. Top 10% of predicted-slow loans against each test year's slowest 20%. Lookup: training-only median by country, sector and month.

1 Better than a simple lookup

Flagging the top 10% gives **2.63–4.04×** lift, against 1.97–2.74× for a historical lookup.

2 Clearest within a country

Median precision is **0.49–0.56**, against 0.22–0.29 for the lookup; random is 0.20.

3 Text adds little to the flag

Adding text changes the tree's precision by **−0.003 to +0.005**.

4 Checked looking back only

Completed loans only; **precision ranges 0.18–0.90** across the largest countries.

→ Why flagged loans come first

Send lender-side support to flagged loans, not story edits

Proposed levers for Kiva and partners; their effect on speed is to be tested, not yet measured

Feature it 1

Show flagged loans to lenders early on the lending page.

Who: kiva

Watch: crowding out other loans 2

Match it

Offer matching funds where a loan is likely to stall.

Who: kiva and funders

Watch: the cost of matching funds

Alert lenders

Notify lenders who follow the loan's sector or country.

Who: kiva

Watch: lender fatigue

Follow up

Ask the field partner to check the listing is complete and clear.

Who: Field partners

Watch: extra partner workload

Don't chase framing tweaks 4

The evidence points away from wording, so effort goes to the loan.

Instead

- Flag slow loans at posting
- Reach the right lenders
- Pilot before any advice





Scope: the levers are proposals; our data do not measure their effect. 3

Independent hackathon analysis; not affiliated with or endorsed by Kiva.

1 Four levers act on lenders

Each changes **who sees a flagged loan** or what support it gets, not the story.

2 Each has a cost to watch

Featuring can crowd out other loans; matching uses money; alerts can **tire lenders**.

3 Their effect is unmeasured

Our data cannot say what these levers do, so **each needs a controlled pilot**.

4 Wording is not the lever

The evidence so far gives **no reason to rewrite stories** to speed funding.

→ Why writing advice must pass gates first

Writing advice has to pass three gates first

Proposed next research for Kiva and participating partners; each gate must pass before the next

1 Validate the measures

Question 1
Do our word counts capture clarity, need and agency?

Do
Build a **small codebook**; **two people** code the same narratives and resolve disagreements.

Done when
Definitions are **frozen** before later periods are checked.

2 Close the data gaps

Question 2
What did lenders see, and which listings never funded?

Do
Get **original texts**, partner identifiers, exposure data and **expired listings**.

Done when
Unfunded listings and **listing-time text** are in the data.

3 Pilot one prompt

Question 3
Does editing a description change funding speed?

Do
With partners, test **one truthful clarity prompt** against a **control**; urgency in Asia is a candidate.

Done when
Threshold, power and guardrails are **agreed in advance**.

Each gate unlocks a stronger claim

Today
Associations and forecast information

1 **After gate 1**
Measures that mean what we say

2 **After gate 2**
Results for every listing, not only funded ones

3 **After gate 3**
A tested effect of editing

If a gate fails, stop there: report the uncertainty and give no writing advice. 4

1 Check the measures first

Word counts may not capture clarity, need or agency until **people have coded them**.

2 Then fill what we cannot see

We lack **what lenders saw** and the listings that never funded.

3 Only a pilot can test an edit

One truthful prompt against a control, with **the pass mark set in advance**.

4 A failed gate stops the work

The claim stays where it is, and **no writing advice goes out**.

→ Why the limits stay on the page

Four limits decide what the solution can claim today

What this evidence cannot show, and what to do about it

Limit	What we cannot show	What we do about it
Associations, not causes 1 — Limit	Every result is observational; no edit or lever has been tested.	**Pilot each lever and prompt against a control.**
Funded loans only 2 — Limit	Listings that expired unfunded are not in the data, so reach is unknown.	**Add expired listings before judging reach.**
Uneven across countries 3 — Check	Flag precision varies widely within the largest countries, and probabilities drift in 2025.	**Recalibrate and validate the flag country by country.**
For whom and when 4 — Checked	Framings stay small across groups, and the one strict-test pass holds in one model only.	**Test urgency in Asia as a hypothesis.**

Where each limit gets addressed

Flag rollout validated by country (3) → **Gate 1** measures checked → **Gate 2** missing data added (2) → **Gate 3** one pilot run (1, 4)

1 Claims stay at association level

We say **linked with**, not caused by, until a pilot has run.

2 Reach stays an open question

We cannot yet say a lever helps **loans that would otherwise expire**.

3 The flag is validated by country

No country gets a flag **until it is recalibrated** on local data.

4 One passing signal is not a rule

It holds in one model only, so **no writing rule follows**.

→ Why the roadmap starts with a pilot

Start with prediction and support now; earn writing advice later

Proposed roadmap for Kiva and partners

Now

1. Pilot the slow-funding flag at listing
2. Route lender-side support to flagged loans
3. Stop investing in framing tweaks

Next

1. Recalibrate and validate the flag by country
2. Validate the text measures with a codebook
3. Pilot one truthful clarity prompt against a control

Later

1. Data partnership: listing-time text, expired listings, partner identifiers
2. Answer for whom and when properly
3. Give writing advice only if the pilot shows it helps

If a step fails, stop there: report the uncertainty and give no writing advice.

Thank you

Mike Nguyen and Sophia Nguyen, Cultural Blend

Code and evidence: github.com/tuannm3812/unsw-ma-hackathon-2026



Regional family links look strong but rest on two countries each

Explanatory models: within-region slopes with country-clustered and heuristic few-cluster intervals



Countries behind each region 3

Africa	27	Central America	2
Asia	12	Middle East	2
Oceania	4	North America	1

Figure A1. Within-region family-framing slopes, 95% ranges, explanatory models; one dot per country, Kiva regions.
[1] t(1) reference on the same clustered error: a heuristic screen, not a calibrated test. Axis in log points × 100.

1 Both links look strong at first

Standard errors put family framing at about **7% faster funding** per extra mention in each region.

2 Two countries cannot carry them

A few-country check widens both ranges **across zero**; the Middle East is **97% Palestine**.

3 Other regions have many more countries

Africa has **27 countries** and Asia **12**; these two regions have two each.

4 No regional writing rules yet

Treat both as **descriptive patterns**; a real test needs more countries or partner-level data.

Each forecast trains only on loans completed before its cutoff

Forecast benchmark: training records must have completed before each cutoff

Evaluation year	Pre-cutoff postings completing later	Eligible as-of training	Evaluation rows
2021	7,528	717,672	163,166
2022	3,918	884,448	143,354
2023	3,546	1,028,174	143,233
2024	5,691[1]	1,169,262	141,725
2025	4,261	1,312,417	137,162

Figure A2. Training availability and evaluation rows by year.

[1] A posting-only split would keep these 5,691 training records; evaluation windows still select eventual completions.

The primary models hold up on later, completed loans

Explanatory models: chronological holdout, completed loans only

5.2031 days

duration MAE, HistGradientBoosting

0.5420

duration R^2

0.8997

24-hour ROC AUC

0.8301

24-hour average precision

0.1156

24-hour Brier score

Scope: 1,174,953 training and 278,887 holdout loans (2024–2025). Retrospective predictions on completed loans, not deployment estimates.

Story themes track loan purpose more than writing style

Exploratory NMF themes from a 20,000-loan sample; descriptive, not full-data evidence

1.5 days

mean funding time, fastest theme (sanitation)

13.5 days

mean funding time, slowest theme (group solar and farm plots)

What the themes show

8 themes, including sanitation, clean water, livestock, smallholder farming and retail. They track loan purpose as much as writing style.

How to read them

Means are unadjusted and come from a sample, so the gap is context, not a tested effect of story content.

Text lowers a linear model's error; a stronger tree barely changes

Forecast benchmark: linear model, structure only versus structure plus text, identical evaluation rows

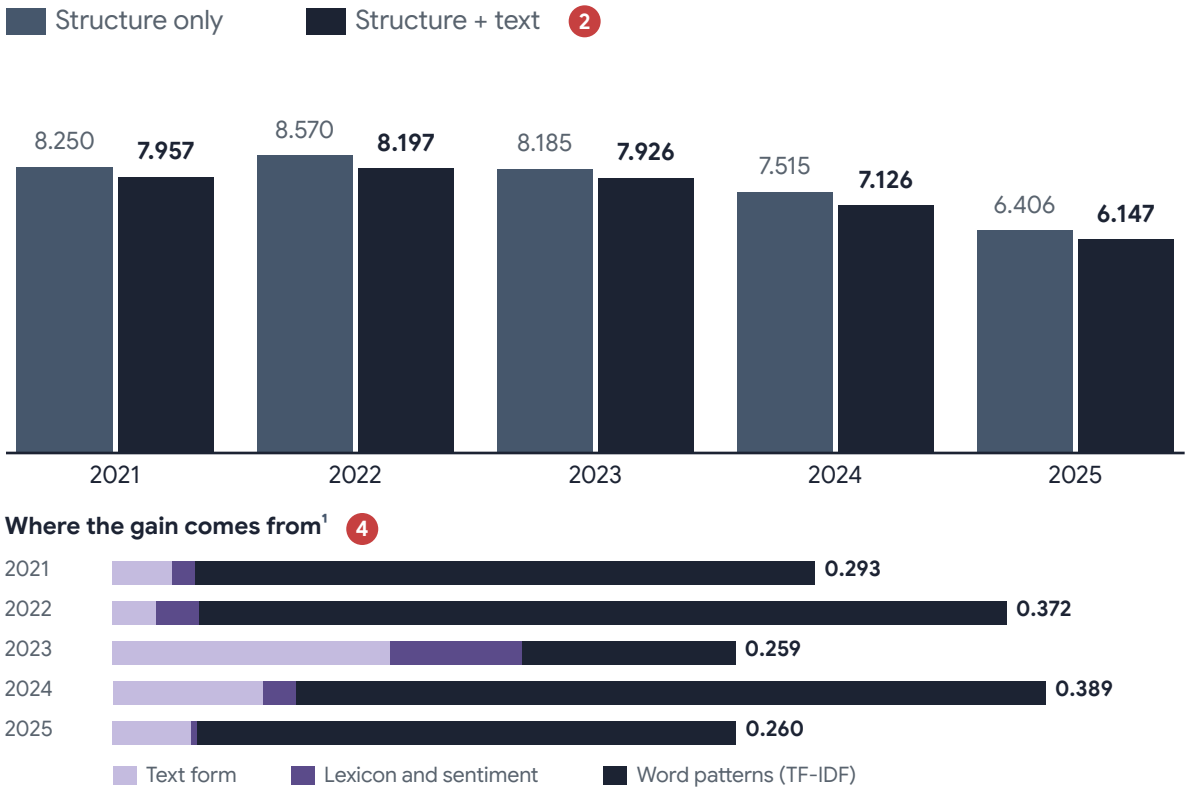


Figure A5. Linear-model MAE in days (top) and its reduction by text feature group (bottom), identical evaluation rows.

[1] Groups added in order: form, then lexicon and sentiment, then TF-IDF; another order would split the gain differently.

0.29 days 1

median annual error reduction

5 of 5 3

years with lower error

728,640

evaluation records

1 A small linear gain

Gains of **0.26–0.39 days** on 728,640 evaluation records; about half is context.

2 Forecasting is not persuasion

Text may encode **purpose, place or programme templates**, not only persuasion.

3 No single country drives it

Removing any single country **leaves the gain positive** in every year (exploratory check).

4 Word patterns carry most of it

TF-IDF gives the largest share in **4 of 5 years**; lexicon and sentiment add only **0.003–0.055 days**.

Text became numbers through classical NLP, not a large language model

Every text feature in the deck, how it was built, and where its limits are

Technique	What it captures	Settings	Where used	Limit
Description form	Length and structure of the text	Characters, words, sentences, average word and sentence length	Explanatory models and forecasts	Says nothing about content
Word-list counts	Eight framing measures, including first and third person	Theory-guided lists of 4–13 entries; rate per 100 words	Explanatory, robustness and equivalence; forecasts	Not yet validated (gate 1); counts words, not meaning
VADER sentiment	Positive or negative tone	Rule-based lexicon; compound score from –1 to 1	Explanatory, robustness and equivalence; forecasts	General-purpose lexicon, not built for loan text
TF-IDF word patterns	Distinctive words and two-word phrases	Up to 5,000 features, in at least 5 training loans; fitted on training years only	Linear forecast	May pick up place and programme templates
Compressed text features	TF-IDF reduced for the boosted tree	Truncated SVD, 64 components; fitted on training years only	Larger boosted tree	Hard to interpret
NMF topics	Eight story themes	TF-IDF up to 1,000 features; 20,000-loan sample	Appendix A4 only	Exploratory; sample only, not full-data evidence

Figure A6. Methods as implemented in src/ and the exploratory notebook. No large language model or pretrained embeddings were used.